1. Kubernetes Setup: Set up a local Kubernetes cluster using Minikube

1.1 Introduction

Kubernetes is an open-source container orchestration platform used to automate the deployment, scaling, and management of containerized applications.

Kubernetes helps manage containers efficiently by organizing them into clusters and managing resources such as pods, services, and deployments.

In this assignment, a **local Kubernetes cluster was created using Minikube**, and a simple application was deployed to demonstrate Kubernetes resource management.

1.2 Tools Used

Tool	Purpose
Kubernetes	Container orchestration
Minikube	Local Kubernetes cluster
kubectl	Kubernetes CLI
Helm	Kubernetes package manager
Docker	Container runtime

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2. Kubernetes Setup using Minikube

2.1 Install Minikube

Download and install Minikube.

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
sudo install minikube-linux-amd64 /usr/local/bin/minikube
```

Verify installation:

minikube version

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$
```

2.2 Install kubectl

Install Kubernetes CLI.

sudo apt install kubectl -y

Verify installation:

kubectl version --client

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$ kubectl version --client
Client Version: v1.35.1
Kustomize Version: v5.7.1
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$
```

2.3 Start Kubernetes Cluster

Start Minikube cluster.

minikube start

Verify cluster:

kubectl get nodes

Expected output:

NAME STATUS ROLES
minikube Ready control-plane

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   29m   v1.35.1
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$
```

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3. Application Deployment

A simple Nginx application was deployed to the Kubernetes cluster.

3.1 Create Deployment

Create deployment:

kubectl create deployment nginx-deployment --image=nginx

Verify deployment:

kubectl get deployments

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
my-nginx           1/1     1            1           9m7s
nginx-deployment   1/1     1            1           27m
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$
```

3.2 Check Pods

Pods are the smallest deployable units in Kubernetes.

kubectl get pods

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6ff797d4c9-vg5bn   1/1     Running   0          3m12s
```

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4. Resource Management

4.1 Expose Deployment using Service

Expose deployment:

kubectl expose deployment nginx-deployment --type=NodePort --port=80

Check service:

kubectl get services

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$ kubectl get svc
NAME               TYPE           CLUSTER-IP     EXTERNAL-IP   PORT(S)                    AGE
kubernetes         ClusterIP      10.96.0.1      <none>        443/TCP                    27m
my-nginx           LoadBalancer   10.99.143.69   <pending>     80:30590/TCP,443:31018/TCP 7m19s
nginx-deployment   NodePort       10.99.162.57   <none>        80:32218/TCP               25m
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$
```

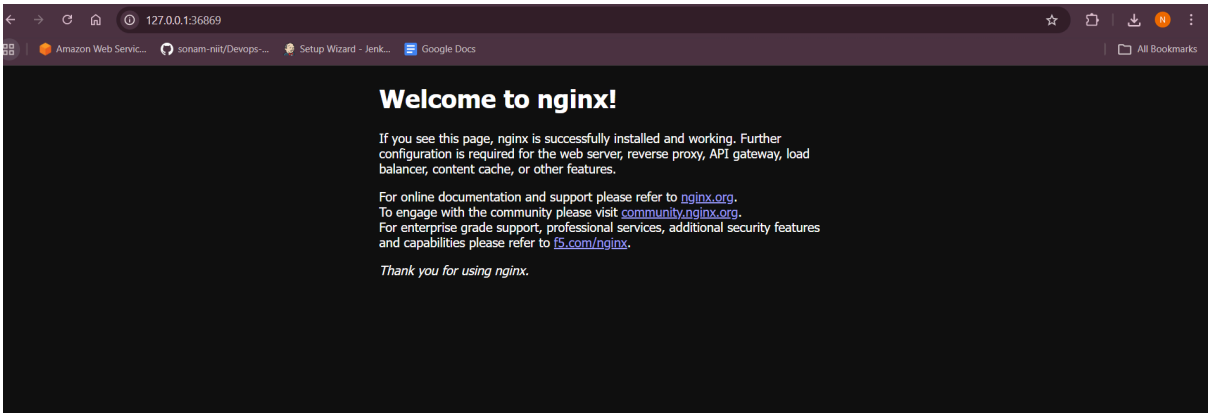
4.2 Access Application

Open service URL:

minikube service nginx-deployment

This command opens the application in the browser.

📸 **Screenshot**



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5. Managing Kubernetes Resources

Common Kubernetes commands:

Resource	Command
Pods	kubectl get pods
Deployments	kubectl get deployments
Services	kubectl get services

Example:

kubectl get all

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$ kubectl get all
NAME                                     READY   STATUS    RESTARTS   AGE
pod/nginx-deployment-6ff797d4c9-vg5bn    1/1     Running   0          3m22s

NAME                       TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
service/kubernetes         ClusterIP   10.96.0.1      <none>        443/TCP        4m52s
service/nginx-deployment   NodePort    10.99.162.57   <none>        80:32218/TCP   2m35s

NAME                               READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/nginx-deployment   1/1     1            1           3m23s

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/nginx-deployment-6ff797d4c9   1         1         1       3m23s
```

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6. Helm Charts

Helm is a package manager for Kubernetes that simplifies application deployment.

6.1 Install Helm

curl https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash

Verify installation:

helm version

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$ helm version
version.BuildInfo{Version:"v3.20.0", GitCommit:"b2e4314fa0f229a1de7b4c981273f61d69ee5a59", GitTreeState:"clean", GoVersion:"go1.25.6"}
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$
```

6.2 Add Helm Repository

helm repo add bitnami https://charts.bitnami.com/bitnami
helm repo update

6.3 Deploy Application using Helm

Install Nginx using Helm:

helm install my-nginx bitnami/nginx

Verify:

helm list

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$ helm list
NAME            NAMESPACE       REVISION        UPDATED                                   STATUS          CHART           APP VERSION
my-nginx        default         1               2026-03-12 09:22:49.581900721 +0000 UTC deployed        nginx-22.6.4    1.29.6
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments$
```

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7. Advantages of Kubernetes

- Automated container orchestration
- Self-healing containers
- Automatic scaling
- Load balancing
- Efficient resource utilization

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8. Conclusion

In this assignment, a Kubernetes cluster was successfully set up using Minikube. A simple Nginx application was deployed and managed using Kubernetes resources such as deployments, pods, and services. Helm was also used to simplify application deployment.

This demonstrates how Kubernetes simplifies container orchestration and application management.